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I. INTRODUCTION

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II. RELATED WORK

A. Egocentric Body Motion Estimation

Recent research on egocentric body motion estimation has made significant advances in accuracy and efficiency. FRAME [1] utilizes geometric multimodal integration and floor-aligned representations to achieve state-of-the-art body pose prediction with real-time performance at 300 FPS. REWIND [2] employs cascaded body-hand denoising diffusion and exemplar-based identity conditioning to accomplish real-time high-fidelity whole-body motion estimation. Specifically, Wang et al. [3] use FisheyeViT and a diffusion-based motion refinement approach to achieve accurate and temporally stable whole-body motion capture from a single head-mounted fisheye camera. Dai et al. [4] employ a multi-modal data collection strategy with LiDAR and IMUs to create the SLOPER4D dataset, facilitating global 4D human pose estimation in large-scale urban environments. Wang et al. [5] utilize a spatio-temporal optimization framework integrated with learned motion priors to accomplish egocentric 3D human pose estimation in global coordinates, addressing issues like temporal jitter and occlusion. Akada et al. [6] introduced the UnrealEgo dataset and a multi-branch autoencoder-based model to estimate 3D human poses from egocentric stereo videos, addressing challenges like self-occlusion with a heatmap-driven approach. Building on this, Akada et al. [7] proposed a transformer-based framework that integrates scene depth features and temporal query augmentation, accompanied by the UnrealEgo2 dataset, to enhance pose estimation accuracy in dynamic environments. EgoPoseFormer [8] introduces a transformer-based model for stereo egocentric

3D human pose estimation, which innovates through a two-stage framework comprising a Pose Proposal Network for coarse joint localization and a Pose Refinement Transformer with Deformable Stereo Attention to leverage multi-view features for accurate pose refinement.

B. Body Motion Estimation

Research on 3D human body motion estimation aims to reconstruct 3D joint coordinates from raw image data or estimated 2D joint positions, typically through monocular or multi-view inputs. KTPFormer [9] introduces a kinematics and trajectory prior knowledge-enhanced transformer that incorporates graph convolutional networks to explicitly model anatomical constraints and motion trajectories for more physically plausible 3D human pose estimation. MotionAGFormer [10] proposes an adaptive graph attention transformer that dynamically learns joint relationships through multi-scale graph convolutions and temporal attention to capture both structural dependencies and motion patterns. TCPFormer [11] presents a novel temporal correlation learning framework using implicit pose proxies as intermediate representations, enabling comprehensive temporal modeling through proxy update, invocation, and attention modules for sequence-to-sequence pose estimation. GraphMLP [12] introduces a graph-reinforced MLP-like architecture that combines multi-layer perceptrons (MLPs) and graph convolutional networks (GCNs) in a unified framework to capture global, local, and graphical interactions for 3D human pose estimation. Cheng et al. [13] propose a deep-learning framework with temporal convolutional networks (TCNs) and a Cylinder Man Model to explicitly handle occlusions by filtering unreliable keypoints and enforcing temporal smoothness in video-based 3D pose estimation. PyMAF-X [14] presents a pyramidal mesh alignment feedback (PyMAF) loop and an adaptive integration strategy with elbow-twist compensation to achieve well-aligned full-body model regression from monocular images by leveraging multi-scale spatial features.

III. METHOD

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A. Learnable Graph Convolution Module

To address the unique challenges of egocentric pose estimation in XR environments, we propose an enhanced framework that incorporates a learnable graph convolution module to explicitly model anatomical relationships between human joints. This approach is particularly relevant for wearable device scenarios where frequent occlusions and limited field of view pose significant challenges for accurate pose estimation.

The core component of our enhanced architecture is a learnable graph convolution module that operates directly on 3D joint coordinates. Given input joint features $X \in \mathbb{R}^{B \times T \times V \times J \times 3}$, where B is batch size, T is temporal length, V is number of views, and J is number of joints, the graph convolution is formulated as:

$$\mathbf{X}_{transformed} = \mathbf{X}\mathbf{W} \quad (1)$$

$$\hat{\mathbf{A}} = softmax\left(\frac{\mathbf{A} + \mathbf{A}^T}{2}\right) \quad (2)$$

$$\mathbf{X}_{output} = \hat{\mathbf{A}}\mathbf{X}_{transformed} + \mathbf{b} \quad (3)$$

where $\mathbf{W} \in \mathbb{R}^{3 \times 3}$ is the learnable weight matrix initialized using Xavier uniform initialization, $\mathbf{A}$ is the learnable adjacency matrix.

The graph convolution module is integrated into the STF architecture through a residual connection with learnable weighting:

$$\mathbf{X}_{graph} = LayerNorm(GraphConv(\mathbf{X}_{in})) \quad (4)$$

$$\mathbf{X}_{enhanced} = \mathbf{X}_{in} + \lambda \cdot \mathbf{X}_{graph} \quad (5)$$

where λ is a learnable residual weight parameter that starts at 0.1 and adapts during training to balance the contribution of graph-enhanced features.

The module processes input sequences by first enhancing joint representations with learned anatomical constraints, then passing these graph-refined features to subsequent spatio-temporal modeling components. This approach allows the model to leverage both explicit anatomical knowledge and data-driven temporal patterns for robust pose estimation in challenging egocentric scenarios.

The pipeline design demonstrates that explicit modeling of joint relationships at the coordinate level, combined with careful integration strategies, provides significant advantages for egocentric applications where traditional methods struggle with partial observability and dynamic viewing conditions.

B. Implicit Joint Localization Module

Accurate 3D human pose estimation from monocular images remains challenging due to the inherent depth ambiguity and complex spatial relationships. Traditional approaches often rely on explicit 2D joint detection as an intermediate step, which can introduce error propagation and suboptimal performance. To address these limitations, we propose an Implicit Joint Localization (IJL) module that learns joint-aware representations without explicit 2D coordinate regression during inference.

The IJL module employs a novel teacher-forcing strategy with scheduled sampling during training, where 2D ground truth coordinates serve as supervision signals for attention mechanism learning rather than as direct inputs. This approach enables the model to develop implicit joint localization capabilities while maintaining end-to-end differentiability.

Given an input feature map $F \in \mathbb{R}^{C \times H \times W}$ extracted from the backbone network, we first generate Gaussian target heatmaps for each joint using the ground truth 2D coordinates during training. For each joint k with normalized coordinates (x_k, y_k) , we compute a Gaussian heatmap $G_k \in \mathbb{R}^{(H \times W)}$ as follows:

$$G(i, j) = exp\left(-\frac{(i - y^2) + (j - x^2)}{2\sigma^2}\right) \quad (6)$$

where (i, j) denotes the spatial coordinates in the heatmap, and σ controls the spread of the Gaussian distribution. This formulation creates a probabilistic representation of joint locations, with higher values indicating greater likelihood of joint presence.

The core of our IJL module is an attention mechanism that learns to localize joints implicitly. We generate attention maps $A_k \in \mathbb{R}^{H \times W}$ for each joint through a lightweight convolutional layer:

$$A = \text{sigmoid}(\text{Conv}_{1 \times 1}(F)) \quad (7)$$

where $A_k \in \mathbb{R}^{K \times H \times W}$ represents the attention maps for all K joints. During training, we supervise the attention maps using the Gaussian target heatmaps through an attention loss function:

$$\mathcal{L}_{attention} = \frac{1}{K} \sum_{k=1}^K \|A_k - G_k\|_2^2 \quad (8)$$

This loss function encourages the network to learn attention maps that approximate the Gaussian distributions centered at the true joint locations.

Once the attention maps are learned, we aggregate joint-specific features through spatial weighting. For each joint k , we compute the weighted feature vector $f_k \in \mathbb{R}^C$ as:

$$f_k = \sum_{i=1}^H \sum_{j=1}^W A_k(i, j) \cdot F(i, j) \quad (9)$$

The final joint feature representation is obtained by stacking all joint features:

$$F_{joints} = [f_1, f_2, \dots, f_K]^T \in \mathbb{R}^{K \times C} \quad (10)$$

This representation captures both the appearance information and spatial relationships crucial for accurate 3D pose estimation.

C. Adaptive Spatio-Temporal Fusion Gate (ASTFG) Module

To address the limitations of fixed feature fusion strategies in existing spatio-temporal transformers for 3D human pose estimation, we propose an Adaptive Spatio-Temporal Fusion Gate (ASTFG) module. Traditional approaches often employ simple addition or concatenation operations to combine spatial and temporal features, which may not optimally capture the dynamic dependencies between joint relationships and motion patterns. Our key insight is that the relative importance of spatial configuration versus temporal dynamics varies significantly across different actions and motion phases.

The ASTFG module introduces a content-aware gating mechanism that dynamically adjusts the contribution weights of spatial and temporal feature streams. This adaptive fusion strategy enables the model to emphasize more informative feature components based on the input characteristics, leading to more robust 3D pose estimation, particularly in challenging scenarios with occlusions or rapid movements.

Let $F_s \in \mathbb{R}^{B \times N \times C}$ represent the spatial features extracted by the Kinematic Priori Attention (KPA) block and $F_t \in \mathbb{R}^{B \times N \times C}$ denote the temporal features from the Temporal Priori Attention (TPA) block, where B is the batch size, N is the number of joints, and C is the feature dimension.

The fusion process begins by computing global context vectors through average pooling:

$$\mathbf{g}_s = \frac{1}{N} \sum_{i=1}^N F_s^{(i)}, \quad \mathbf{g}_t = \frac{1}{N} \sum_{i=1}^N F_t^{(i)} \quad (11)$$

where $\mathbf{g}_s, \mathbf{g}_t \in \mathbb{R}^{B \times C}$ represent the global spatial and temporal context, respectively. These context vectors are concatenated and processed by a lightweight gating network to generate adaptive fusion weights:

$$\mathbf{z} = [\mathbf{g}_s; \mathbf{g}_t] \in \mathbb{R}^{B \times 2C} \quad (12)$$

$$\mathbf{w} = \text{Softmax}(\mathbf{W}_2 \cdot \text{ReLU}(\mathbf{W}_1 \cdot \mathbf{z} + \mathbf{b}_1) + \mathbf{b}_2) \quad (13)$$

where $\mathbf{W}_1 \in \mathbb{R}^{\frac{B}{c} \times 2C}$, $\mathbf{W}_2 \in \mathbb{R}^{2 \times \frac{B}{c}}$ are learnable parameters, r is the reduction ratio, and $\mathbf{w} = [w_s, w_t] \in \mathbb{R}^{B \times 2}$ contains the adaptive weights for spatial and temporal features. The final fused feature F_{fused} is computed as:

$$\mathbf{F}_{fused} = w_s \cdot \mathbf{F}_s + w_t \cdot \mathbf{F}_t + \mathbf{B} \quad (14)$$

where $\mathbf{B} \in \mathbb{R}^{1 \times 1 \times C}$ is a learnable bias term that enhances representational capacity.

D. Figures and Tables

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TABLE I
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IV. EXPERIMENTS

A. Dataset

The FRAME dataset captures extensive multimodal data from 14 subjects, each performing two sessions with varied attire to enhance visual diversity. Each session begins with a brief calibration phase, followed by 50 scripted actions (e.g., sports and daily routines) lasting 20 seconds with short pauses, ensuring a wide range of motion patterns. Participants interacted naturally with their environment using VR headsets equipped with forward-facing cameras. The dataset includes stereo egocentric videos, 6D device poses, markerless body motion captures, and multi-view recordings from a 120-camera studio, totaling over 7 hours of data (1.6 million images). A separate test set with two additional subjects provides 1 hour of data for unbiased evaluation. Key ground-truth annotations per frame include: dual fisheye images $I_t \in \mathbb{R}^{2 \times 3 \times H \times W}$, device poses $T_D(t) \in SE(3)$ (aligned to the floor), and 3D joint positions $J_t \in \mathbb{R}^{J \times 3}$. This large-scale, richly annotated dataset supports robust training and testing for egocentric 3D pose estimation tasks.

B. Metrics

As in prior works, we report results in terms of mean per joint positional error (MPJPE) in millimeters. In addition, we report other complementary metrics, i.e., Procrustes-aligned MPJPE (PA-MPJPE), 3D percentage of correct keypoints (3D-PCK) within a 10cm threshold, Jitter, Non-Penetration Percentage (NPP), Mean Penetration Error (MPE), and Foot Sliding (FS). A detailed explanation on how these metrics are computed is present in the supplementary materials.

C. Implementation Details

D. Experimental Results

ACKNOWLEDGMENT

The preferred spelling of the word “acknowledgment” in America is without an “e” after the “g”. Avoid the stilted expression “one of us (R. B. G.) thanks ...”. Instead, try “R. B. G. thanks...”. Put sponsor acknowledgments in the unnumbered footnote on the first page.

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